



Genome-wide identification of Dicer-like, Argonaute and RNA-dependent RNA polymerase gene families and their expression analyses in response to viral infection and abiotic stresses in *Solanum lycopersicum*

Miao Bai ^{a,1}, Guo-Shun Yang ^{a,1}, Wen-Ting Chen ^a, Zhen-Chuan Mao ^{b,c}, Hou-Xiang Kang ^{b,c}, Guo-Hua Chen ^b, Yu-Hong Yang ^{b,c}, Bing-Yan Xie ^{b,c,*}

^a Key Laboratory for Crop Germplasm Innovation and Utilization of Hunan Province, Hunan Agricultural University, Changsha 410128, China

^b Institute of Vegetables and Flowers, Chinese Academy of Agricultural Sciences, Beijing 100081, China

^c Key Laboratory of Biology and Genetic Improvement of Horticultural Crops, Ministry of Agriculture, China

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ABSTRACT

Dicer, Argonaute and RNA-dependent RNA polymerase form the core components to trigger RNA silencing. Although tomato (*Solanum lycopersicum*) is a dicotyledon model plant, no systematic analysis and expression profiling of these genes in tomato has been undertaken previously. In this study, seven Dicer-like (SIDCLs), 15 Argonaute (SIAGOs) and six RNA-dependent RNA polymerase (SIRDs) genes were identified in tomato. These genes were categorized into four subgroups based on phylogenetic analyses. Comprehensive analyses of gene structure, genomic localization and similarity among these genes were performed. Their expression patterns were investigated by means of expression models in different tissues and organs using online data and semi-quantitative RT-PCR. Many of the candidate genes were up-regulated in response to Tomato yellow leaf curl virus infection and abiotic stresses. The expression models of tandem gene duplications among SIDCL2s indicated the DCL2 family plays an important role in the evolution of tomato.

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1. Introduction

RNA silencing is guided by numerous kinds of small silencing RNAs, such as microRNAs (miRNAs), short-interfering RNAs (siRNAs) and PIWI-interacting RNAs (piRNAs) (Ding, 2010). It is widely implicated as an important molecular mechanism to tune the expression of many genes in eukaryotic organisms. In plants, RNA silencing pathways are triggered by small RNAs, which are usually mediated by the interaction of Dicer-like proteins (DCLs), Argonautes (AGOs) and RNA-dependent RNA polymerases (RDRs). These proteins are implicated in the initiation and maintenance of the mechanism central to

this mode of gene regulation (Kapoor et al., 2008). Most RNA silencing pathways in plants can be divided into three critical stages, which are initiation, maintenance and signal amplification. First, different sources of double-stranded RNAs (dsRNAs), such as miRNA precursors, hairpin RNAs, viral RNAs or transgene loci (Deleris et al., 2006; Fusaro et al., 2006; Gascoli et al., 2005; Lund, 2004; Xie et al., 2005; Yoshikawa et al., 2005) are processed by the RNaseIII-type activities of DCLs into small RNAs. Generally, one strand of these small RNAs is then loaded into the RNA-induced silencing complex (RISC), which contains AGOs. Under participation of AGOs with sequences complementary to the small RNAs, silencing then occurs by either cleavage or blocking translation of the target mRNA. In the signal amplification stage, RDR enzymes are responsible for synthesis of dsRNAs using the RNA as the template, and then the dsRNAs are cut into secondary small RNAs by DCL and give rise to a new round of RNA silencing.

Recently, the structures and functions of these core proteins have been clarified. For example, six types of domains, viz., DEAD-helicase, helicase-C, Duf283, PAZ, RNaseIII and double stranded RNA-binding (dsRB) domains, are characteristic in Dicers (Margis et al., 2006). Argonaute proteins contain several functional domains, including DUF1785, PAZ, MID and PIWI domains, as revealed by structural studies (Hutvagner and Simard, 2008), whereas RDR proteins contain a unique RNA-dependent RNA polymerase (RdRP) domain.

Multiple copies of Dicers, AGO and RDR genes exist in eukaryotes. In plants, four DCL, 10 AGO and six RDR genes of *Arabidopsis* and a total of

Abbreviations: AGO, Argonaute; CS, chilling stress; CDS, coding DNA sequence; CK, control; DCL, Dicer-like protein; dsRNAs, double-stranded RNAs; DS, drought stress; HS, heat stress; hc-siRNAs, heterochromatic siRNAs; ITAG, International Tomato Annotation Group; miRNAs, microRNAs; MS, Murashige and Skoog medium; OS, oxidative stress; piRNAs, PIWI-interacting RNAs; PTGS, post-transcriptional gene silencing; qPCR, Quantitative reverse transcription PCR; RT-PCR, reverse transcription PCR; RDR, RNA-dependent RNA polymerase; RdRP, RNA-dependent RNA polymerase domain; RISC, RNA-induced silencing complex; SA, salicylic acid; SS, salt stress; siRNAs, short-interfering RNAs; SMART, Simple Modular Architecture Research Tool; TYLCV, tomato yellow leaf curl virus; TYLCD, tomato yellow leaf curl disease.

* Corresponding author at: Institute of Vegetables and Flowers, Chinese Academy of Agricultural Sciences, 12 Zhongguancun Nan Avenue, Haidian, Beijing 100081, China. Tel.: +86 10 82109545; fax: +86 10 62174123.

E-mail address: xiebingyan2003@yahoo.com.cn (B.-Y. Xie).

¹ The first two authors contributed equally to this paper.

32 genes in rice and 28 genes in maize have been identified (Kapoor et al., 2008; Qian et al., 2011). Function analysis of these genes revealed that different genes play multiple roles in regulating growth and development as well as in response to abiotic and biotic stress.

Tomato (*Solanum lycopersicum*) is an important vegetable crop and a model plant for the research of fruit development and plant defense (Kimura and Sinha, 2008), specifically to study virus-derived/induced RNA silencing and plant systemic gene silencing. At present, although many years have passed since the first plant RDR was identified from tomato, most other RNA silencing-related genes are unclear. A project to sequence the tomato genome was launched by the International Solanaceae Project (SOL) (Mueller et al., 2005, 2009). Now, the genome sequence data VERSION 2.3 on the assembled draft genome sequence of *S. lycopersicum* is available from (http://solgenomics.net/genomes/Solanum_lycopersicum/genome_data.pl). The genome sequence data can provide a basis to identify RNA silencing-associated genes in tomato.

2. Materials and methods

2.1. Identification of candidate genes

Amino acid sequences of all the DCL, AGO, and RDR genes of *Arabidopsis* and rice were downloaded from the National Center for Biotechnology Information (NCBI) database and conserved domains were searched using the Pfam protein families database (Pfam 24.0) (Finn, 2011). HMMER 3.0 software (<http://hmm.janelia.org/>) was employed to predict for DCL, AGO and RDR genes encoded in the tomato genome sequence data VERSION 2.3. We combined the Blast tool (<http://solgenomics.net/tools/bblast/index.pl>) search results generated from the Tomato genome build release SL 2.31, Nov. 2010 (International Tomato Genome Sequencing Consortium; <http://solgenomics.net/>) to identify each gene. Meanwhile, the conserved domains were checked by Pfam 24.0 program. Finally, Lycopersicon (all tomato species) SGN mRNA sequences (by Blast tool in <http://solgenomics.net/>) and the Micro-Tom Database (<http://www.pgb.kazusa.or.jp/mibase/>) (Aoki, 2010) were used to analyze gene expression in silico. No expressed sequence tag (EST) supported genes were detected by RT-PCR. All obtained sequences were collected in Candidate Protein/CDS Group A (Supplementary material S1). Compute pI/Mw (http://web.expasy.org/compute_pi/) was used to calculate the proteins' isoelectric point (PI) and molecular weight (Mw). Introns were located in International Tomato Annotation Group (ITAG) using GBrowse. The Pfam database and Simple Modular Architecture Research Tool (SMART) were adopted to search and annotate conserved domains in complete amino acid sequences of DCL, AGO and RDR genes in tomato.

2.2. Phylogenetic analysis, chromosomal localization and similarity analysis of genes

For phylogenetic analysis, total protein sequences of Candidate protein group A and these query proteins sequences (Supplementary material S1) were aligned using ClustalW in MEGA5 (Tamura et al., 2007). The neighbor-joining (NJ) method was used to construct a phylogenetic tree. The stability of tree nodes was tested by bootstrap analysis with 1000 replicates. The number of amino acid substitutions between sequences (pairwise distances) was also computed using MEGA5. Candidate proteins in group A were named based on the nomenclature used for previously identified genes and on the basis of their phylogenetic relatedness (according to the pairwise distances) to other members of the same family. A chromosomal location image for candidate genes was generated according to the genomic browse data on tomato genome sequence data. Because the tomato genome is not fully assembled, gene locus information may need to be updated. Gene similarity matrixes were generated by DNAMAN with the multiple sequence alignment method.

2.3. Plant material and stress treatment

Tomato (*S. lycopersicum* 'Moneymaker') plants were grown in a greenhouse at 25 ± 2 °C with a photoperiod of 16 h. Different organs (shoot, leaf, flower, fruit and root) were harvested (as listed in Table S2: Supplementary material S2) from 2-month-old plants. To investigate gene expression in response to virus stress, an infectious clone of tomato yellow leaf curl virus TYLCV-[CN:SH2] (AM282874) (Zhang et al., 2009) was agro-inoculated into 5-week-old tomato plants, which were then cultured in a greenhouse for 28 d. After checking that the infectious clone had been successfully inoculated by PCR with TYLCV-[CN:SH2] specific primers (Supplementary material S3) and typical symptoms had emerged, we harvested leaves showing typical symptoms of yellow leaf curl and healthy leaves of non-inoculated plants as a control. For abiotic stress treatments, 14-day-old tomato seedlings were subjected to drought (dried on tissue paper while other abiotic stresses were steeped in liquid Murashige and Skoog [MS] medium), salt (200 mM NaCl), oxidative (100 μ M methyl viologen; Sigma, USA), chilling (4 °C) and heat (40 °C) stress for 3 h. Non-treated seedlings immersed in liquid MS medium for 3 h were used as a control. Three biological replicates for each treatment were harvested for RNA isolation. After harvesting, samples were frozen in liquid nitrogen and stored at -70 °C. Serial numbers and symbols used are listed in Table S2.

2.4. RNA isolation and RT-PCR analysis

RNA was isolated using TRIzol reagent (Invitrogen, USA) according to the manufacturer's protocol and genomic DNA was removed with DNase I (Invitrogen) treatment. First-strand cDNA was synthesized from 2 μ g total RNA using the SuperScript™ III cDNA Synthesis Kit (Invitrogen) according to the manufacturer's instructions. All genes were subjected to semi-quantitative RT-PCR using specific primers (Supplementary material S3), which were designed by Primer Premier 5. Tomato *actin* (BT013707) was used as an internal control for normalization. The reaction profile used was as follows: 94 °C for 4 min, 30 cycles of 94 °C for 30 s, 60–62 °C (Supplementary material S3) for 40 s, 72 °C for 40 s, and final elongation at 72 °C for 5 min. Each PCR pattern was verified by triple replicate experiments, reaction mixture without template was used as a negative control and with *actin* as a positive control. A 10 μ L aliquot of the reaction was separated on 1.5% agarose gel. As a control, reactions were run in parallel that excluded reverse transcriptase. Quantitative RT-PCR (qPCR) was carried out using Bio-Rad CFX96 Real-Time system and Bio-Rad CFX Manager software (version 1.5) (Bio-Rad, USA). The primers used are listed in Supplementary material S3. Tomato *actin* gene was used as an internal control for normalization. At least three replicates of each cDNA sample were performed for qPCR analysis.

3. Results

3.1. Identification and structural analysis of tomato DCL, AGO and RDR genes

We first obtained data for all DCL, AGO and RDR genes in NCBI, which had been investigated in *Arabidopsis* and rice. The query sequences are included in Supplementary material S1. Based on these query sequences, HMM and Blast tool analyses, we identified seven genes encoding Dicer-like proteins (SIDCLs), 15 encoding Argonautes (SIAGO) and six RDR genes (SIRDR) in the tomato genome database (Table 1).

The seven newly identified Dicer-like loci showed coding potentials of 1317 to 1888 amino acid polypeptides. *SIDCL1* (Solyc10g005130) was characterized by the longest open reading frame of 5667 bp with a coding potential for 1888 amino acid polypeptides. The highest number of introns among SIDCLs was 20, the maximum number of introns (23)

Table 1
Identification of tomato DCL, AGO and RDR genes.

Gene name	Accession numbers ¹	Chromosome location	ORF length (bp)	No. of introns	Protein			Evidence ²
					Length (aa)	PI	Mw (Da)	
<i>DICER-like</i>								
1 <i>SIDCL1</i>	Solyc10g005130	ch10 112143–125039	5667	20	1888	6.02	211 925.72	A B
2 <i>SIDCL2a</i>	Solyc06g048960	ch06 28550915–28562223	4200	21	1399	6.46	157 865.48	A B
3 <i>SIDCL2b</i>	Solyc11g008540	ch11 2715282–2724508	4059	20	1352	6.30	152 476.42	A B
4 <i>SIDCL2c</i>	Solyc11g008520	ch11 2690602–2697409	4287	20	1428	7.45	161 274.11	A B
5 <i>SIDCL2d</i>	Solyc11g008530	ch11 2700331–2706663	3954	20	1317	7.48	149 396.49	A B
6 <i>SIDCL3</i>	Solyc08g067210	ch08 53434291–53462298	4293	18	1430	6.50	160 379.56	A B
7 <i>SIDCL4</i>	Solyc07g005030	ch07 53267–79984	4611	23	1536	6.40	173 341.16	A B
<i>Argonautes</i>								
8 <i>SIAGO1a</i>	Solyc06g072300	ch06 40962769–40969607	3165	20	1054	9.51	116 855.84	A B C
9 <i>SIAGO1b</i>	Solyc03g098280	ch03 54076103–54082377	2943	21	980	9.25	109 211.67	A B
10 <i>SIAGO10a</i>	Solyc09g082830	ch09 63889312–63895465	2949	20	982	9.30	110 522.00	A B
11 <i>SIAGO10b</i>	Solyc12g006790	ch12 1201257–1207114	2802	21	933	9.29	105 494.43	A B
12 <i>SIAGO5</i>	Solyc06g074730	ch06 42663263–42669576	3036	21	1011	9.41	112 641.38	A B
13 <i>SIAGO7</i>	Solyc01g010970	ch01 6612318–6616163	3003	2	1000	9.27	114 006.88	B
14 <i>SIAGO2a</i>	Solyc02g069260	ch02 33781267–33787661	3129	2	1042	9.39	116 000.23	A B
15 <i>SIAGO2b</i>	Solyc02g069270	ch02 33791034–33794941	2934	3	977	9.15	109 979.25	B
16 <i>SIAGO3</i>	Solyc02g069280	ch02 33797120–33800704	3000	2	999	8.91	112 672.26	A B
17 <i>SIAGO6</i>	Solyc07g049500	ch07 57095518–57105721	2712	21	903	8.75	101 553.25	A B
18 <i>SIAGO4a</i>	Solyc01g008960	ch01 2978867–2984620	2730	21	909	8.97	101 752.29	A B C
19 <i>SIAGO4b</i>	Solyc06g073540	ch06 41727326–41732814	2742	21	913	9.01	101 880.14	A B
20 <i>SIAGO4c</i>	Solyc06g073530	ch06 41718888–41723862	2715	21	904	9.00	101 278.68	N
21 <i>SIAGO4d</i>	Solyc01g096750	ch01 79536534–79541185	2646	21	881	9.26	98 779.06	N
22 <i>SIAGO15</i>	Solyc03g111760	ch03 56447333–56452967	2751	21	916	9.58	104 684.50	N
<i>RNA-dependent RNA polymerases</i>								
23 <i>SIRDR1</i>	Solyc05g007510	ch05 2062518–2069664	3345	3	1114	8.45	126 808.10	A B C D
24 <i>SIRDR2</i>	Solyc03g114140	ch03 58205446–58210730	3360	3	1119	6.91	127 369.71	A B
25 <i>SIRDR6a</i>	Solyc04g014870	ch04 5125732–5129867	3594	1	1197	6.39	136 227.07	A B
26 <i>SIRDR6b</i>	Solyc08g075820	ch08 57079477–57083527	3540	2	1179	7.69	133 581.15	B
27 <i>SIRDR3a</i>	Solyc12g008410	ch12 1847925–1857522	2493	15	830	8.26	95 228.30	A B C
28 <i>SIRDR3b</i>	Solyc06g051170	ch06 30850530–30862377	1728	11	575	7.56	65 572.60	A B

1, Accession numbers correspond to annotations provided by ITAG2.3. 2, Source of gene expression evidence: A, ESTs database; B, segment RT-PCR; C, full-length CDS; D, previous research (Schiebel et al., 1998); N, no evidence. The RT-PCR fragment sequences of *SIDCL2c*, *SIAGO7*, *SIAGO2b*, *SIAGO3*, *SIRDR6b* and *SIRDR3b* that have no or little EST data support were submitted to ESTdb of NCBI (GenBank accession numbers are JK006277 to JK006282, see Supplementary material S1).

was found in *SIDCL4*. Only partial of full-length sequences of DCL genes was present in the SGN mRNA sequences database, except for related EST data and RT-PCR evidence (Table 1). Pfam and SMART analyses revealed that *SIDCL1*, *SIDCL2d* and *SIDCL4* contained DEAD, Helicase-C, DUF283, PAZ, RNaseIII and dsRB domains, which are typical structures of the Dicer family (class 3 RNaseIII family) (MacRae and Doudna, 2007). Both DCL1 and DCL4 clades (see Fig. 1) contained two RNaseIII and dsRB domains, but the DCL2 clade (e.g., *AtDCL2*) contained only a dsRB domain, while *SIDCL2a*, *b* and *c* lacked a dsRB domain. Genes in the DCL3 clade coded for an incompletely known protein structure; *AtDCL3* lacked the DUF238 domain and *SIDCL3* lacked the DEAD domain, and both contained only one incomplete dsRB domain (Fig. 1).

The 15 Argonautes coded for ~100 kDa basic proteins with pI ranging from 8.75 to 9.58. These SIAGOs were characterized by the presence of PAZ and PIWI domains, which typically configure plant AGO proteins. Most of the SIAGO genes included about 21 introns, but *SIAGO7*, *SIAGO2a*, *SIAGO2b* and *SIAGO3* contained only two or three introns. Among the SIAGO genes, no EST data existed to support *SIAGO7*, *SIAGO2b*, *SIAGO4c*, *SIAGO4d* and *SIAGO15*, but RT-PCR fragments demonstrated the presence of *SIAGO7* and *SIAGO2b*. The additional SIAGO (named *SIAGO4a*) was found to correspond to the tomato mRNA sequence (BT013764) in this study. Unlike *SIAGO4a* and *4b*, we detected no specific amplified fragments of *SIAGO4c*, *SIAGO4d* and *SIAGO15* in 30 cycles. For these three genes, which have no corresponding EST data, no transcripts could be detected in any organ or under any stress analyzed (see below), even after more than 35 cycles or with an increased quantity of cDNA template (data not shown). These genes might be expressed during specific vegetative and reproductive development stages or might just be pseudogenes. This requires a more detailed investigation.

Consistent with the typical structure of the plant AGO family, DUF1785, PAZ and PIWI domains were detected in all of the 15 SIAGOs (Fig. 1). In addition, Gly-rich Ago1 domains, which are conserved in *AtAGO1*, were found in *SIAGO1a* and *1b* but were absent in the other AGOs. Previous research revealed that the PIWI domain of AGOs adopts a folded structure similar to that of RNase H enzymes and exhibits endonuclease activity owing to an active site usually carrying an aspartate–aspartate–histidine (DDH) motif and/or histidine at position 798 (H798) (Baumberger and Baulcombe, 2005; Kapoor et al., 2008). To interrogate whether SIAGOs possessed the conserved catalytic residues and could potentially act as the slicer component of RISC, we aligned the PIWI domains of all of the SIAGOs (Fig. 3). Six proteins (*SIAGO1a*, *SIAGO1b*, *SIAGO10a*, *SIAGO10b*, *SIAGO5* and *SIAGO7*) possessed the conserved DDH/H798 residues. D760 and D854 in tomato and *Arabidopsis* were conserved except in *SIAGO15*. Similarly, H986 was also conserved in most tomato SIAGOs, but it was replaced by aspartate in *SIAGO2a*, *SIAGO2b*, *SIAGO3*, *AtAGO2* and *AtAGO3*, which were grouped in the ZIPPY/AGO7 clade. Interestingly, compared to previous studies that showed a variety of H798 in monocots (Kapoor et al., 2008; Qian et al., 2011), H798 sites in the AGO4 group of tomato (*SIAGO4a*, *b*, *c*, *d* and *SIAGO6*) were all replaced by proline (Table 2).

The first RNA-directed RNA polymerase was isolated from tomato (Schiebel et al., 1998). All of the six RDR genes present in the tomato genome encoded proteins that share a common RdRP domain. The full-lengths of these candidate SIRDRs varied from 1728 bp for *SIRDR3b* to 3594 bp for *SIRDR6a*, with coding potentials of 575 and 1197 amino acids, respectively. Interestingly, the genomic sequence of *SIRDR6a* that encodes the longest open reading frame is interrupted by a single intron, while *SIRDR3a* (locus Solyc12g008410) encodes an

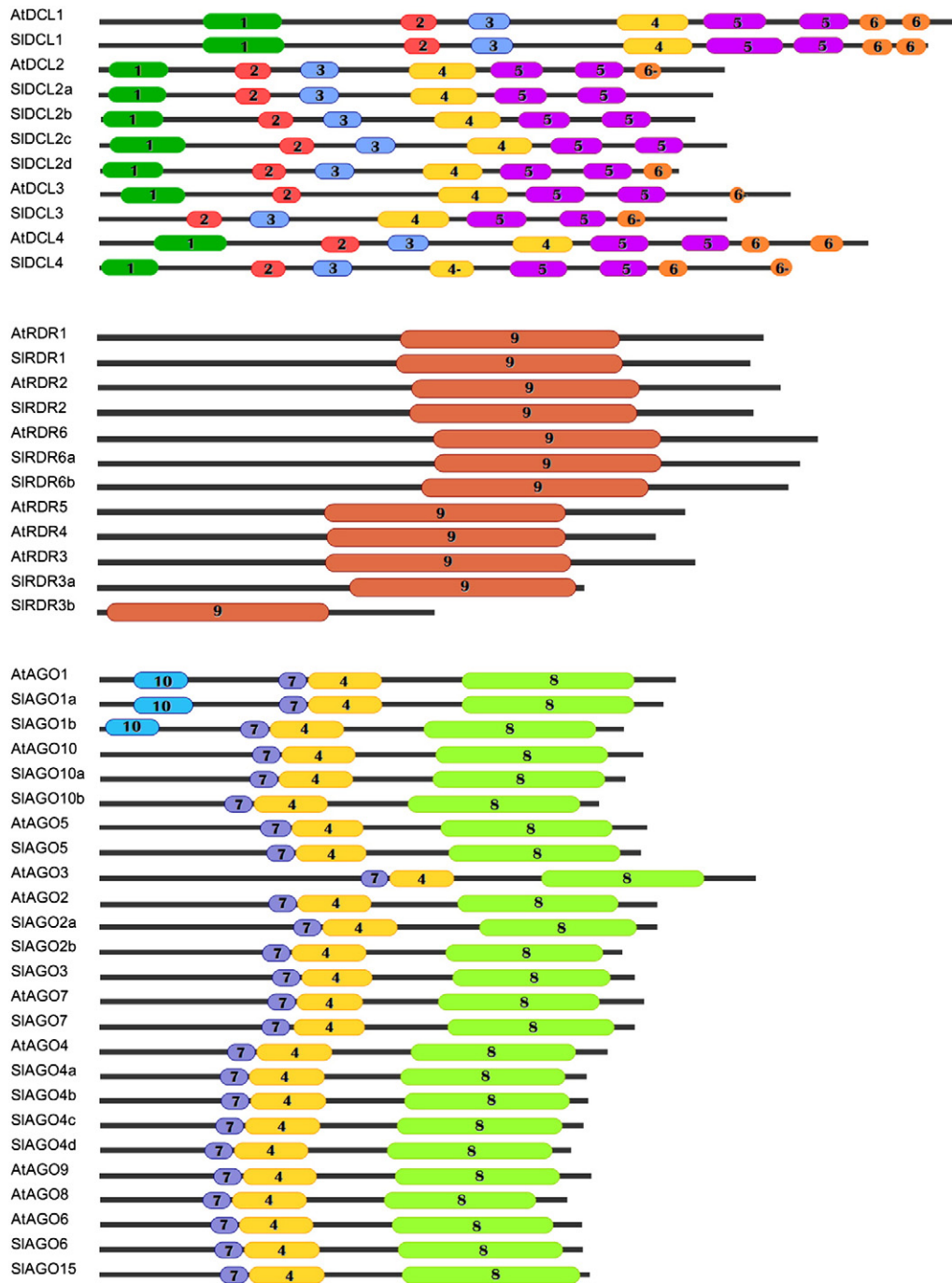


Fig. 1. Predicted peptide domain analysis of DCL, AGO and RDR genes. 1: DEAD domain; 2: helicase-C domain; 3: Duf283 domain; 4: PAZ domain; 5: RNaseIII domain; 6: double stranded RNA-binding (dsRB) domain; 7: DUF1785 domain; 8: PIWI domain; 9: RdRP domain; 10: Gly-rich AGO1 domain; Incomplete domains are indicated with “-”.

open reading frame of 2970 bp containing 15 introns (Table 1). The six tomato RDR genes contained a unique RdRP domain, whereas *SIRDR3a* and *SIRDR3b* had a shorter N terminal and C terminal, respectively (Fig. 1). More detailed site information of these domains analyses are provided in Supplementary material S4.

3.2. Phylogenetic analysis of plant DCL, AGO and RDR genes

To determine evolutionary relatedness of tomato AGO, RDR and DCL with those of *Arabidopsis* and rice, total protein sequences from these organisms were aligned and an unrooted neighbor-joining tree was constructed (Fig. 2). Based on these analyses, seven candidate SIDCLs were divided into four groups and the respective genes

were designated as *SIDCL1*, *SIDCL2a*, *SIDCL2b*, *SIDCL2c*, *SIDCL2d*, *SIDCL3* and *SIDCL4* (Fig. 2A). In the DCL2 clade, similarities among the four DCLs proteins ranged from 78.3% to 89.0% (Supplementary material S5); the genes were named *SIDCL2a*, *b*, *c* and *d*, respectively. By contrast, the other three subfamilies contained only one newly identified SIDCL and were named after their counterparts in *Arabidopsis* and rice based on sequence similarity.

The unrooted neighbor-joining tree generated from aligned full-length protein sequences of all 15 SIAGO, 10 AtAGO and 19 OsAGO proteins was resolved into four clades, namely AGO1, MEL1/AGO5, ZIPPY/AGO7 and AGO4 (Fig. 2B). In clade 1, four tomato genes were designated as *SIAGO1a*, *SIAGO1b*, *SIAGO10a* and *SIAGO10b* based on their high sequence homologies with *AtAGO1* and *AtAGO10*. A second

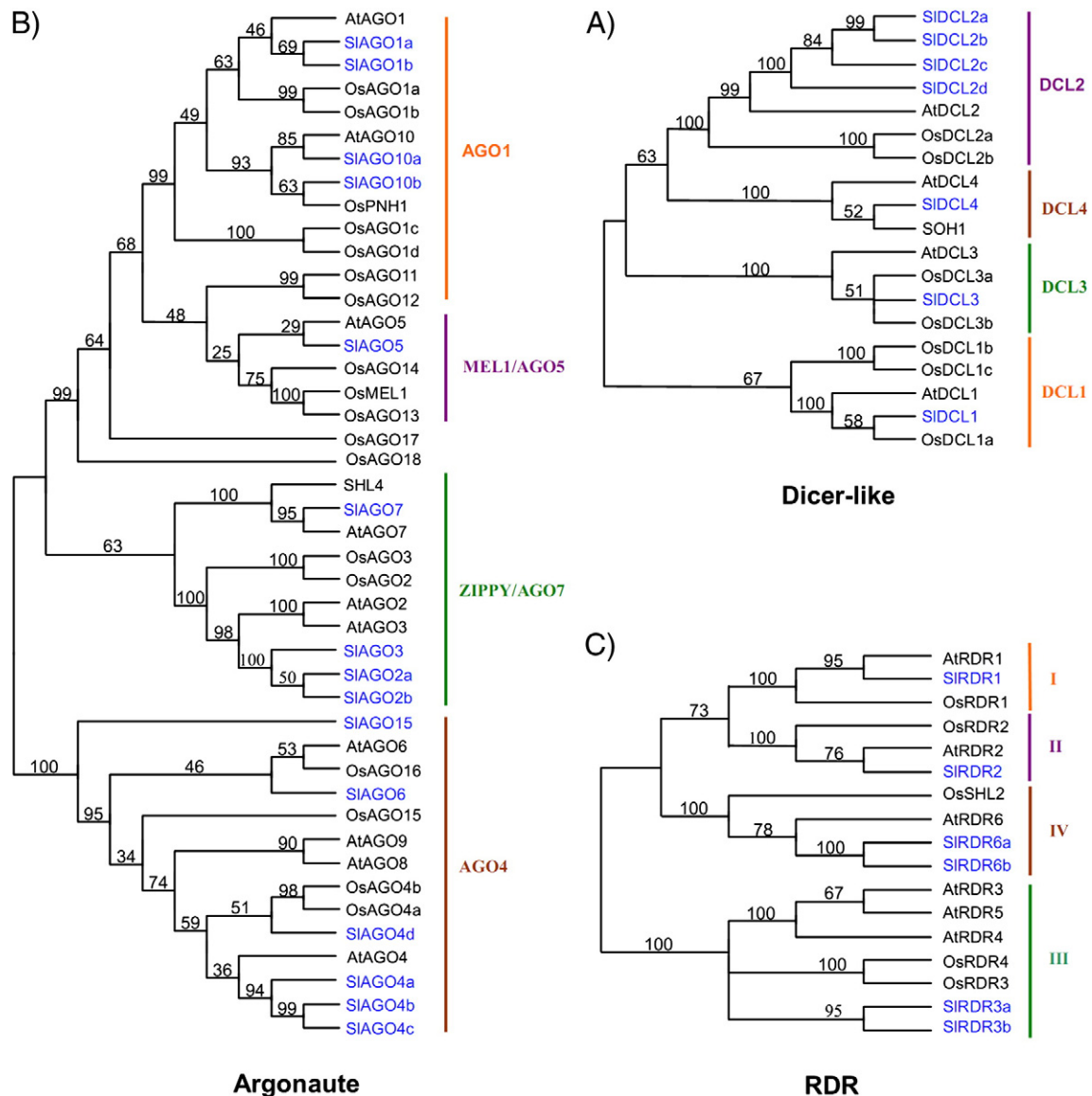


Fig. 2. Unrooted neighbor-joining trees constructed from multiple alignments of total (A) DCL, (B) AGO or (C) RDR protein sequences of tomato, *Arabidopsis* and rice. Bootstrap support values from 1000 replications are indicated above the branches. Each gene family is divided into four clades (Kapoor et al., 2008) shaded with a different color. Tomato genes are highlighted in blue. Sequences of *Arabidopsis* and rice were downloaded from the NCBI database (see Supplementary material S1 for accession numbers).

AGO clade comprised proteins similar to *AtAGO5* and contained only one tomato protein, which was named *SIAGO5*. Four tomato proteins clustered with three *Arabidopsis* proteins (*AtAGO2*, *AtAGO3* and *AtAGO7*) and three rice proteins (*OsAGO2*, *OsAGO3* and *SHL4*) in the ZIPPY/AGO7 subfamily and the respective genes were designated as *SIAGO2a*, *SIAGO2b*, *SIAGO3*, and *SIAGO7* according to their high sequence similarities with *AtAGO2*, 3 and 7, respectively. The AGO4 subfamily contained 14 proteins, among which six tomato genes were designated as *SIAGO6*, *SIAGO4a*, *SIAGO4b*, *SIAGO4c*, *SIAGO4d* and *SIAGO15*. In this clade, *SIAGO4a*, *b*, *c* and *d* were similar to *AtAGO4* showing high sequence similarities ranging from 65.5% to 86.1% (Supplementary material S5). No tomato AGO displayed a high similarity with *OsAGO17* and *OsAGO18*.

The neighbor-joining tree derived from RDR protein sequences was resolved into four clades (Fig. 2C). Tomato RDR (Solyc05g007510) was, in previous research (Schiebel et al., 1998), named *SIRDR1* in accordance with the high sequence similarity to *AtRDR1*. The genes for the other five tomato RDRs were named *SIRDR2*, *SIRDR3a*, *SIRDR3b*, *SIRDR6a* and *SIRDR6b* based on their similarities with *Arabidopsis* RDRs.

3.3. Chromosomal localization and similarity analyses of tomato DCL, AGO and RDR genes

To investigate the evolution of multiple DCLs, AGOs and RDRs in tomato, we analyzed their genomic distribution by localizing genes on tomato chromosomes (Fig. 4) and performed multiple sequence alignment to calculate the similarity among each gene (detailed in Supplementary material S5). A total of seven SIDCL genes were distributed on five chromosomes. Three SIDCL genes (*SIDCL2b*, *SIDCL2c* and *SIDCL2d*) were distributed on chromosome 11, while the remainders were located on chromosomes 6, 7, 8 and 10 and represented a single SIDCL. As described above, alignment analyses of the protein sequences showed that there were high similarities (78.3% to 89.0%) among the four tomato DCLs (*SIDCL2a*, *b*, *c* and *d*). According to the principle of Gu et al. (2002), *SIDCL2a* could represent a duplicated gene derived from other *SIDCL2*s, but in rice Kapoor (2008) identified a number of high-similarity genes that were not located in duplicated segments of the genome and cannot be considered the result of gene duplication. In tomato, no data on duplicated segments of the genome

<i>AtAGO1</i>	728	KINVK	-----	-----	-----	VGGRTNVLVD	ALS	RRIPL	VSDRPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KEWDPQKGV
<i>SIAGO1a</i>	730	KINVK	-----	-----	-----	VGGRTNVLVD	ALS	RRIPL	VSDRPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>SIAGO1b</i>	656	KINVK	-----	-----	-----	VGGRTNVLVD	ALS	RRIPL	VSDRPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO10a</i>	675	KINVK	-----	-----	-----	MGGRNVLVD	ALS	CRIPL	VSDIPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>SIAGO10b</i>	670	KINVK	-----	-----	-----	MGGRNVLVD	ALS	CRIPL	VSDIPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO5</i>	624	KINVK	-----	-----	-----	MGGRNVLVD	ALS	RRIPL	VSDIPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO5</i>	687	KINVK	-----	-----	-----	MGGRNVLVD	ALS	RRIPL	VSDIPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO5</i>	703	KINVK	-----	-----	-----	MGGRNVLVD	ALS	RRIPL	VSDIPTIIFG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO7</i>	698	KINAK	-----	-----	-----	IGGSMTELYN	SIPSH	IPRL	RPDEPVIFMG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO7</i>	701	KINAK	-----	-----	-----	VGGCTVALYN	SLPSQ	IPRL	KHDCPVIFMG	ADVTHHPHGE	DS-SPSIAAV	VASQDWPEIT	KYAGLVCAQA	IRQELIQDLF	KTQDPVTRGT
<i>AtAGO2</i>	715	KINAK	-----	-----	-----	VGGSNVLMQ	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO2a</i>	759	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO2b</i>	697	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO3</i>	893	KINAK	-----	-----	-----	VGGTNVLMQ	N	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO3</i>	708	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO6</i>	589	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO6</i>	604	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO9</i>	598	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO8</i>	570	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>AtAGO4</i>	626	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO4a</i>	610	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO4b</i>	614	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO4c</i>	605	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO4d</i>	582	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----
<i>SIAGO15</i>	611	KINAK	-----	-----	-----	LGGSNMLMD	-----	-----	-----	-----	-----	-----	-----	-----	-----

<i>AtAGO1</i>	VTGGMKELL	IAFRST-GH	KPLRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO1a</i>	VTGGMKELL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO1b</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO10a</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO10b</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO5</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO5</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO7</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO7</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO2</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO2a</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO2b</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO3</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO3</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO6</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO6</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO9</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO8</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>AtAGO4</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO4a</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO4b</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO4c</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO4d</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA
<i>SIAGO15</i>	VSOGGIMDLL	ISPRAT-GQ	KPQRIIFYRD	GVSEGGFYQV	LLYELDAIRK	ACASLEAGYQ	PPVTFVVVQK	RHHTLFAQN	HNDRIH-SVD	RSNGILPGTV	VDSKICHTP	EFDFYLCSHA

<i>AtAGO1</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1050
<i>SIAGO1a</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1054
<i>SIAGO1b</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	980
<i>AtAGO10a</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	988
<i>SIAGO10b</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	982
<i>AtAGO5</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	933
<i>AtAGO5</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	997
<i>AtAGO7</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1011
<i>AtAGO7</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	990
<i>AtAGO2</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1000
<i>SIAGO2a</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1014
<i>SIAGO2b</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1042
<i>AtAGO3</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	977
<i>AtAGO3</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	1194
<i>AtAGO6</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	999
<i>AtAGO6</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	878
<i>AtAGO9</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	903
<i>AtAGO8</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	896
<i>AtAGO4</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	850
<i>SIAGO4a</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	924
<i>SIAGO4b</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	909
<i>SIAGO4c</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	913
<i>SIAGO4d</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	904
<i>SIAGO15</i>	GIQGTSPRAH	YHVLWDENNF	TADGLQSLTN	NLCYTYARCT	RSVSIVPPAY	YALAAFRAR	FYMEPET-SD	SGMSAGSMA	R-GGGMAGR	STRGPNVNA	VRPLPALKEN	VKRVMFYC	881

Fig. 3. Alignment profile of PIWI domain amino acids of tomato and *Arabidopsis* AGO proteins. The beginning and end positions of PIWI domains in each protein are marked. The conserved DDH triad residues corresponding to D760, D845 and H986 of *Arabidopsis* AGO1 are highlighted in red, and the conserved H residue corresponding to *Arabidopsis* AGO1 H798 is highlighted in blue.

Table 2

Missing catalytic residue(s) in PIWI domains of tomato and *Arabidopsis*.

Tomato	Argonaute	Motifs ¹	Arabidopsis	Argonaute	Motifs ¹
<i>SIAGO2a</i>	DDD/H		<i>AtAGO2</i>	DDD/H	
<i>SIAGO2b</i>	DDD/H		<i>AtAGO3</i>	DDD/H	
<i>SIAGO3</i>	DDD/S		<i>AtAGO4</i>	DDH/S	
<i>SIAGO6</i>	DDH/P		<i>AtAGO6</i>	DDH/P	
<i>SIAGO4a</i>	DDH/P		<i>AtAGO8</i>	DDH/P	
<i>SIAGO4b</i>	DDH/P		<i>AtAGO9</i>	DDH/R	
<i>SIAGO4c</i>	DDH/P				
<i>SIAGO4d</i>	DDH/P				
<i>SIAGO15</i>	GQR/P				

1, Motifs correspond to conserved D760, D845, H986/H798 of *Arabidopsis* AGO1; D, aspartate; H, histidine; S, serine; P, proline; G, glycine; Q, glutamine; R, arginine.

is available. Importantly, *SIDCL2b*, *c* and *d* were closely localized together on chromosome 11, thus we hypothesized they represent tandem gene duplications.

Four SIAGOs were detected on chromosome 6, three each on chromosomes 1 and 2, two on chromosome 3, and one each on chromosomes 7, 9 and 12. SIAGOs were not localized on five chromosomes (4, 5, 8, 10 and 11) (Fig. 4). Multiple sequence alignment of the 15 tomato AGO proteins revealed that similarities were higher among SIAGOs belonging to the same group (Supplementary material S5). Of these, *SIAGO1a* and *SIAGO1b* shared significant homology with 88.4% identity; *SIAGO10a* and *SIAGO10b* shared 71.4% identity, and the mean identity of four genes in clade AGO1 was 74.07%. The tandem duplicated genes *SIAGO2a*, *2b* and *3*, for which the genes were localized on

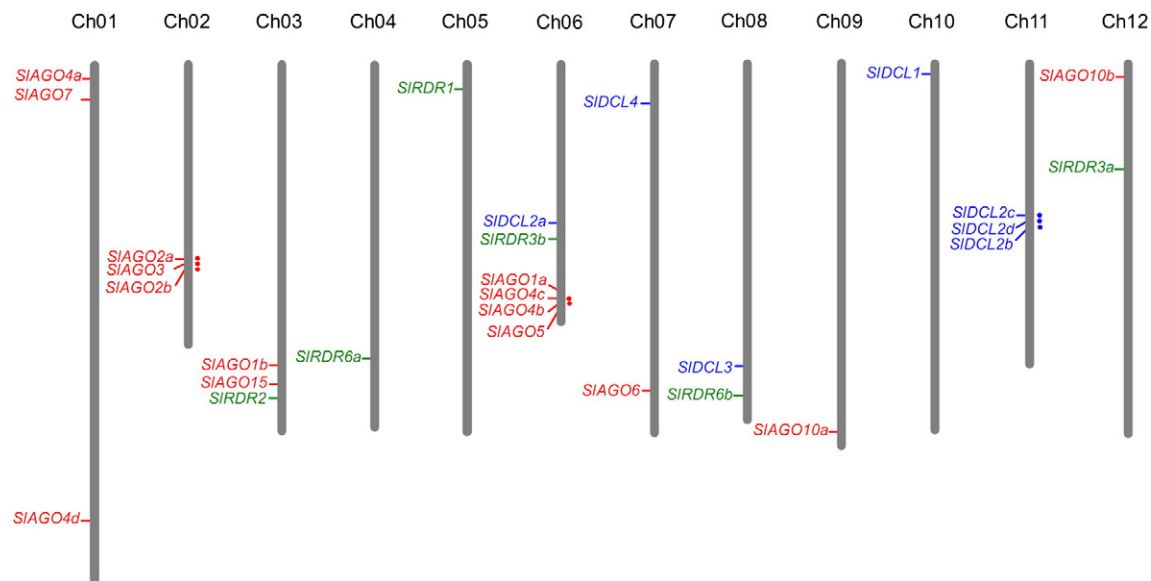


Fig. 4. Chromosomal localization of tomato DCL, AGO and RDR genes. The seven SIDCL, 15 SIAGO and six SIRDR genes were mapped on tomato chromosomes 1 (Ch01) to 12 (Ch12). The chromosome number is specified above each chromosome. Different classified gene families are indicated in different colors. Tandem gene duplications are indicated by filled dots.

chromosome 2, shared high amino acid similarities from 59.8% to 80.1%. In the AGO4 clade, *SIAGO4a*, *b*, *c* and *d* showed about 65% to 86.5% amino acid similarity. Of these, *SIAGO4b* and *c* were localized closely on chromosome 6, and thus were considered to have arisen by tandem duplication.

The six SIRDRs were distributed on six tomato chromosomes (3, 4, 5, 6, 8 and 12), respectively. Except for *SIRDR6a* and *SIRDR6b*, which shared 82.4% similarity, the identities of the other proteins were less than 50%.

3.4. Gene expression models in different tissues and organs

To analyze transcript abundance of these candidate genes in different organs and in response to a variety of stresses in tomato, we performed a comprehensive gene expression in silico analysis using online data and semi-quantitative RT-PCR. The results are shown in Supplementary material S6. Combining these two methods, many of the genes were found to share similar expression patterns. The SGN mRNA and Micro-Tom databases provided a large number of ESTs generated from tomato. However, the online data was incomplete, so the expression of some genes was only detected by semi-quantitative RT-PCR.

Four DCL genes (*SIDCL1*, *2a*, *2b* and *4*), seven AGO genes (*SIAGO1a*, *1b*, *10a*, *10b*, *2a*, *4a* and *4b*), and *SIRDR1* and *2* were expressed in all investigated organs by RT-PCR. Online data showed that most of these genes were detected in many tissues (Table 3). In contrast, most of the other genes (*SIDCL2c*, *SIDCL2d*, *SIDCL3*, *SIAGO5*, *SIAGO6a* and *SIRDR3a*) exhibited moderate expression in the flower or fruit and weak expression in the other organs. However, our data showed that *SIAGO7* and *SIRDR3b* were expressed significantly in the flower, but no or little data supported this finding in silico as was also the case for *SIAGO2b* and *SIRDR6b*. As described above, four *SIAGO4s* of the AGO4 clade shared high sequence identities, but only *SIAGO4a* and *4b* transcripts were detected in tomato. No transcripts of *SIAGO4c*, *SIAGO4d* and *SIAGO15* could be detected in any organ or in response to any stress analyzed.

3.5. Expression of DCL, AGO and RDR genes in response to TYLCV infection and abiotic stresses

When a virus enters a plant cell, viral double-stranded RNA (dsRNA) is targeted by the RNA silencing machinery. Then a cascade

of regulatory events would be initiated, being directed by viral small interfering RNAs (vsRNAs) (Akbergenov et al., 2006; Ding, 2010; Llave, 2010; Wang et al., 2010). Tomato yellow leaf curl virus (TYLCV) is a member of the family Geminiviridae, in which the genome is composed of a circular single stranded DNA. This virus causes tomato yellow leaf curl disease (TYLCD), which affects tomato production greatly and is hard to be managed (Antignus and Cohen, 1994; Czosnek and Laterrot, 1997). Previous studies revealed that RNA silencing is a natural antiviral process and exogenous siRNA could provide resistance to viral infection. But some viruses escape from this mechanism by encoding suppressor proteins (Díaz-Pendón and Ding, 2008; Ding, 2010). Systematic analyses of the RNA silencing system core proteins contribute to revealing the compatible plant-virus interaction.

Semi-quantitative RT-PCR analyses showed that there were significant differences in the expression of related genes between healthy plants and symptomatic plants in response to TYLCV infection (Fig. 5). For example, *SIDCL1*, *2a*, *2c*, *2d* and *3* were significantly up-regulated, but their transcripts were not detected in healthy leaves. Consistent with abiotic stress, most tomato AGOs were up-regulated under TYLCV stress. In addition to *SIRDR2*, *SIRDR6a*, *3a* and *3b* also showed alteration in expression in response to TYLCV infection.

To confirm these predicted genes expression profiles under various stress conditions, five abiotic stress treatments were investigated (Fig. 5). In response to different abiotic stresses, four DCLs (*SIDCL1*, *2a*, *2c* and *2d*) showed up-regulation compared to controls, except for *SIDCL2d* under chilling stress (CS). However, *SIDCL2b*, *SIDCL3* and *SIDCL4* showed approximately equal expression. Three RDR genes (*SIRDR2*, *6a* and *3a*) and five AGO genes (*SIAGO1a*, *1b*, *4a*, *4b* and *5*) showed high expression levels triggered by most of the abiotic stresses. However, many of these genes showed no significant difference between CS and the control (CK). *SIAGO10a* and *10b* were expressed strongly under heat and salt stress conditions, respectively. In this experiment, we found that many genes showed higher expression levels in young leaves (e.g., CK: 2-week-old plant) than mature leaves (e.g., leaf and TYLCV control: 2-month-old plant).

In addition, qPCR was used to obtain insight into some of these gene expression models. Six genes (*SIDCL1*, *SIDCL4*, *SIAGO1a*, *SIAGO4a*, *SIRDR2* and *SIRDR3a*) were represented and further validated by qPCR (Fig. 6). All of these genes were up-regulated to varying degrees by TYLCV infection and abiotic stresses. The transcript accumulation patterns for genes observed by qPCR were similar to those observed

Table 3

Expression of DCL, AGO and RDR genes in different tissues and organs.

	Shoot	Leaf	Flower	Fruit (mixed)	Root	Seed	Seedling	Trichomes	Callus	Crown gall	Unknown	Suspension cultures
1 <i>SIDCL1</i>	P	P	P	P	P	-	S	-	-	-	M	S
2 <i>SIDCL2a</i>	P	P	M S P	P	P	-	-	-	M S	-	-	-
3 <i>SIDCL2b</i>	M S P	M S P	S P	M S P	P	-	S	M	S	MS	MS	-
4 <i>SIDCL2c</i>	-	-	P	P	-	-	-	-	M	-	-	-
5 <i>SIDCL2d</i>	P	P	MP	M S P	P	-	-	MS	MS	-	MS	S
6 <i>SIDCL3</i>	-	-	P	S P	-	-	-	-	S	-	-	-
7 <i>SIDCL4</i>	P	P	M S P	M S P	S P	-	-	-	S	S	MS	S
8 <i>SIAGO1a</i>	M S P	M S P	M S P	P	P	-	-	-	-	MS	-	-
9 <i>SIAGO1b</i>	M S P	M S P	M S P	M S P	M S P	-	-	S	-	MS	-	-
10 <i>SIAGO10a</i>	M S P	S P	M S P	M S P	P	-	-	-	-	-	-	-
11 <i>SIAGO10b</i>	M S P	P	S P	P	P	-	-	-	-	-	-	-
12 <i>SIAGO5</i>	-	-	M P S	P	-	-	-	-	-	MS	MS	-
13 <i>SIAGO7</i>	-	-	P	-	-	-	-	-	-	-	-	-
14 <i>SIAGO2a</i>	P	M S P	P	M S P	S P	-	-	-	MS	S	S	S
15 <i>SIAGO2b</i>	P	-	P	-	-	-	-	-	-	-	-	-
16 <i>SIAGO3</i>	P	P S	P	S P	P	-	-	-	-	-	-	-
17 <i>SIAGO6</i>	P	P	M S P	M S P	P	-	-	-	-	-	MS	S
18 <i>SIAGO4a</i>	M S P	M S P	M S P	M S P	M S P	MS	-	MS	M	-	S	-
19 <i>SIAGO4b</i>	M S P	M S P	M S P	M S P	S P	-	-	S	-	MS	MS	S
20 <i>SIAGO4c</i>	-	-	-	-	-	-	-	-	-	-	-	-
21 <i>SIAGO4d</i>	-	-	-	-	-	-	-	-	-	-	-	-
22 <i>SIAGO15</i>	-	-	-	-	-	-	-	-	-	-	-	-
23 <i>SIRDR1</i>	P	MP	S P	M S P	P	-	S	-	MS	-	MS	S
24 <i>SIRDR2</i>	P	M S P	P	M S P	M S P	-	-	MS	MS	-	MS	S
25 <i>SIRDR6a</i>	MP	P	P	M S P	S	-	-	-	-	-	MS	-
26 <i>SIRDR6b</i>	P	P	P	P	P	-	-	-	-	-	-	-
27 <i>SIRDR3a</i>	P	M S P	MP	M S P	P	-	-	-	-	-	-	-
28 <i>SIRDR3b</i>	-	-	P	S	-	-	-	-	-	-	MS	-

Data were obtained with the Blast tool and downloaded from the following sources: M, Micro-Tom database (<http://www.pgb.kazusa.or.jp/mibase/>); S, *Lycopersicon* (all tomato species) SGN mRNA sequences (<http://solgenomics.net>). In addition, expression in different organs was investigated by semi-quantitative RT-PCR (P; highlighted in orange). RT-PCR results are shown in Supplementary material S6. -, indicates missing data.

from the semi-quantitative RT-PCR analysis, although some showing a significant difference in expression were reflected unclearly in semi-quantitative PCR results. We investigated additional genes for qPCR analysis, but owing to the high CDS sequence identity in the same gene family, the results were often confusing owing to low repeatability or multiple peaks in the melt curve. To solve this problem, we linked all the candidate sequences in the same family to a single line, adopted the Primer Premier 5 filter to screen out highly mismatched primers, and modified manually to obtain ideal primers.

4. Discussion

RNA silencing plays an important role in regulating gene expression at multiple levels during vegetative and reproductive development and is a mechanism for virus resistance and stress tolerance in plants. In recent years, the molecular basis of these complicated and interconnected pathways has been clarified. Concerted activities of DCLs, AGOs and RDRs are required in these processes. However, characteristics and functions of only a few of these genes have been established in tomato. In the present investigation, we identified the genes families in tomato and studied their distribution on chromosomes. Phylogenetic analysis shed light on the evolution of members in each gene family. Semi-quantitative and quantitative RT-PCR-based expression analysis in different organs and under diverse stresses

was attempted to understand the contribution of the activity of these proteins.

4.1. Dicer-like proteins and their multiple roles in tomato

Dicer enzymes function to process dsRNA substrates into small RNA fragments of discrete size. Four DCL genes found in *Arabidopsis* seem to have distinct, hierarchical, and overlapping functions in small RNAs biogenesis (Margis et al., 2006). These proteins catalyze formation of miRNAs and *trans*-acting siRNAs (ta-siRNAs; DCL1 and DCL4), which are well known to trigger post-transcriptional gene silencing (PTGS), 22-nt (DCL2) siRNAs, which share functional overlap in antiviral defense with 21-nt (DCL4) siRNAs, and 24-nt (DCL3) heterochromatic siRNAs (hc-siRNAs) from heterochromatic loci that mediate DNA methylation, gene silencing and chromatin modification (Fukudome et al., 2011; Kurihara and Watanabe, 2004; Qi et al., 2005; Xie, 2004; Xie et al., 2005).

In the present study, seven *SIDCL* genes were identified in the tomato genome. These genes were clustered into four subgroups and among which *SIDCL1* revealed widespread expression in tomato. *SIDCL3* showed low-level expression in vegetative tissues but enhanced transcript accumulation in floral and fruit tissues. The expression pattern was similar for *AtDCL3* and *OsDCL3b*. The similar expression patterns of *SIDCL4*, *SIDCL1* and *SIDCL2a/b* are suggestive of universal

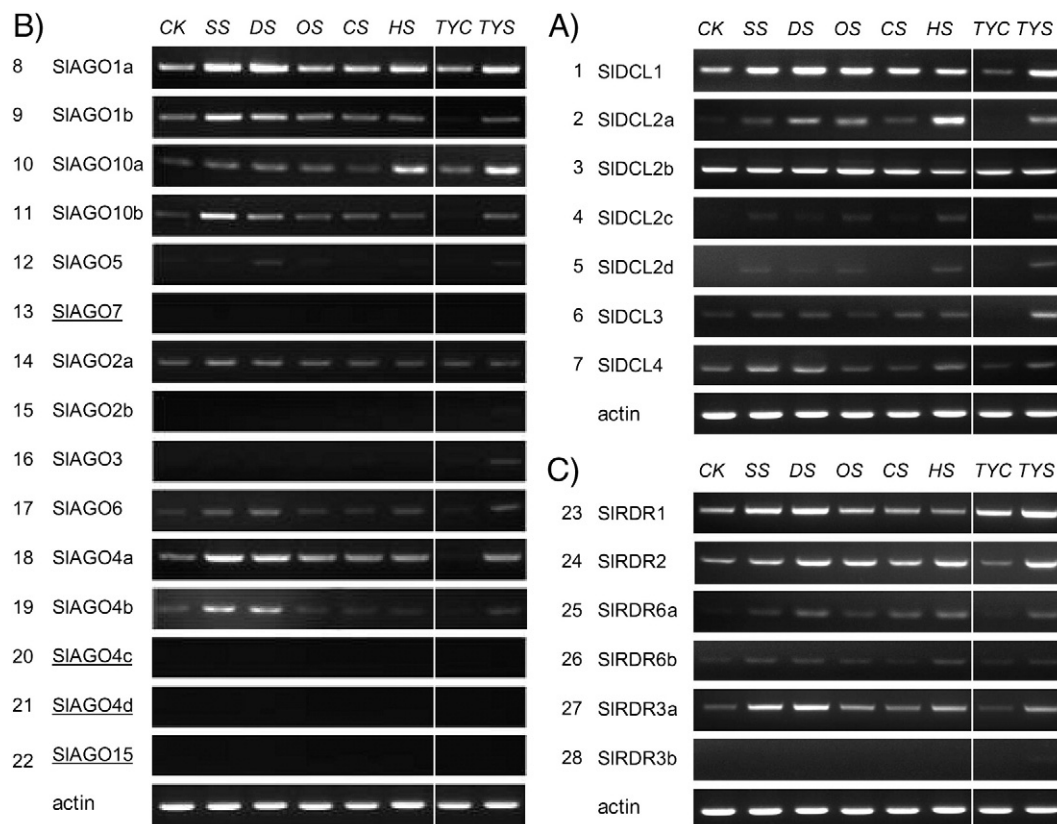


Fig. 5. Semi-quantitative RT-PCR analysis of tomato DCL, AGO and RDR genes under abiotic stress and viral infection treatments. (A) SIDCL, (B) SIAGO and (C) SIRDR genes were detected by semi-quantitative RT-PCR. The cDNA templates were from control (CK), salt stress (SS), drought stress (DS), oxidative stress (OS), chilling stress (CS) and heat stress (HS) treatments; results for TYLCV control (TYC) and TYLCV infection (TYS) are also presented. Tomato actin gene was used as a control for normalization. Gene-specific fragments that were undetected in any template are underlined.

functions for these genes in tomato. In the four *SIDCL* genes (*SIDCL2a*, *b*, *c* and *d*) with high similarity, *SIDCL2b*, *c* and *d* are tandemly duplicated and are located on chromosome 11. Transcripts of these genes were detected in most organs, and *SIDCL2a*, *c* and *d* were shown to be up-regulated in response to TYLCV infection and abiotic stresses. At present, seven DCLs in tomato, six DCLs in poplar and five DCLs in rice and maize have been detected, while only a single species of Dicer known in mammals and nematodes is involved in processing small RNAs. A possible explanation is that, as armed escort, the Dicers were streamlined, because of the presence of the diverse defense system (e.g., immune system and interferons) in mammals. However, plants lack these defense systems, and therefore rely on DCLs to not only regulate their development through miRNAs, but also for defense theme against a multitude of viruses and transposons (Margis et al., 2006).

4.2. Tomato Argonaute genes

Among 10 *Arabidopsis* AGO proteins, a proportion has been functionally associated with different classes of small RNAs (Vaucheret, 2008). *AtAGO1* binds and mediates the functions of most miRNAs, ta-siRNAs, and transgene siRNAs in vivo (Baumberger and Baulcombe, 2005; Fagard et al., 2000; Qi et al., 2005; Vaucheret et al., 2004). *AtAGO1* and *AtAGO10* are most closely related and are both implicated in miRNA-mediated translational repression of target mRNAs (Brodersen et al., 2008; Manavella et al., 2011; Zhu et al., 2011). *AtAGO2* plays a stand-in role for *AtAGO1* in antiviral defense when AGO1 target silencing is overcome by viral suppressors (Harvey et al., 2011). In tomato, expression patterns of four genes of the AGO1 clade indicated that AGO1s may be important for development and defense associated with the small RNAs functions of *Arabidopsis* AGO1 and 10. In addition, whether in tomato development or stress response, *SIAGO2a* expression reflects a

role as the leader of the ZIPPY/AGO7 family. Recent studies revealed that *AtAGO7* preferentially associates with a single miRNA, miR390, to trigger production of ta-siRNAs (Montgomery et al., 2008); AGO5 may be part of a putative germline-specific Argonaute complex association with miRNAs in the *Arabidopsis* male germline (Borges et al., 2011) and *AtAGO9* controls female gamete formation by restricting the specification of gametophyte precursors in a dosage-dependent, non-cell-autonomous manner (Mallory and Vaucheret, 2010). In tomato, *SIAGO5* is highly expressed in the flower and fruit, which is an expression pattern similar to that of *SIAGO6* and *SIAGO7*. This result indicates that these genes may play an important role in reproductive growth of plants and provides a basis for further research into their functions.

4.3. Role of tomato RNA-dependent RNA polymerase

Endogenous eukaryotic RDRs use single-stranded RNAs as templates and generate double-stranded RNA intermediates in diverse processes of small RNAs synthesis in RNA silencing pathways. Among the six *Arabidopsis* RDR genes, *AtRDR1*, 2 and 6 serve in distinct and overlapping processes such as viral resistance, chromatin silencing and PTGS (Diaz-Pendon et al., 2007; Donaire et al., 2008; Kapoor et al., 2008). Of these, *AtRDR1* is elicited by salicylic acid (SA) treatment and viral infection (Diaz-Pendon et al., 2007; Qi et al., 2009; Xie et al., 2001; Yu et al., 2003), while *SIRDR1* is enhanced by salt and drought stress. *AtRDR2* has a crucial role in the biogenesis of hc-siRNAs that induce a DNA methylation pathway with participation of AGO4 and DCL3 (Kasschau et al., 2007; Lu et al., 2006; Raja et al., 2008; Xie, 2004). The homolog of *AtRDR2* in tomato, *SIRDR2*, enhanced expression of closely related AGO4 group members and *SIDCL3* in young plants and during reproductive development, as well as in response to TYLCV infection. These results implied these genes may be involved in the hc-siRNAs collaborative

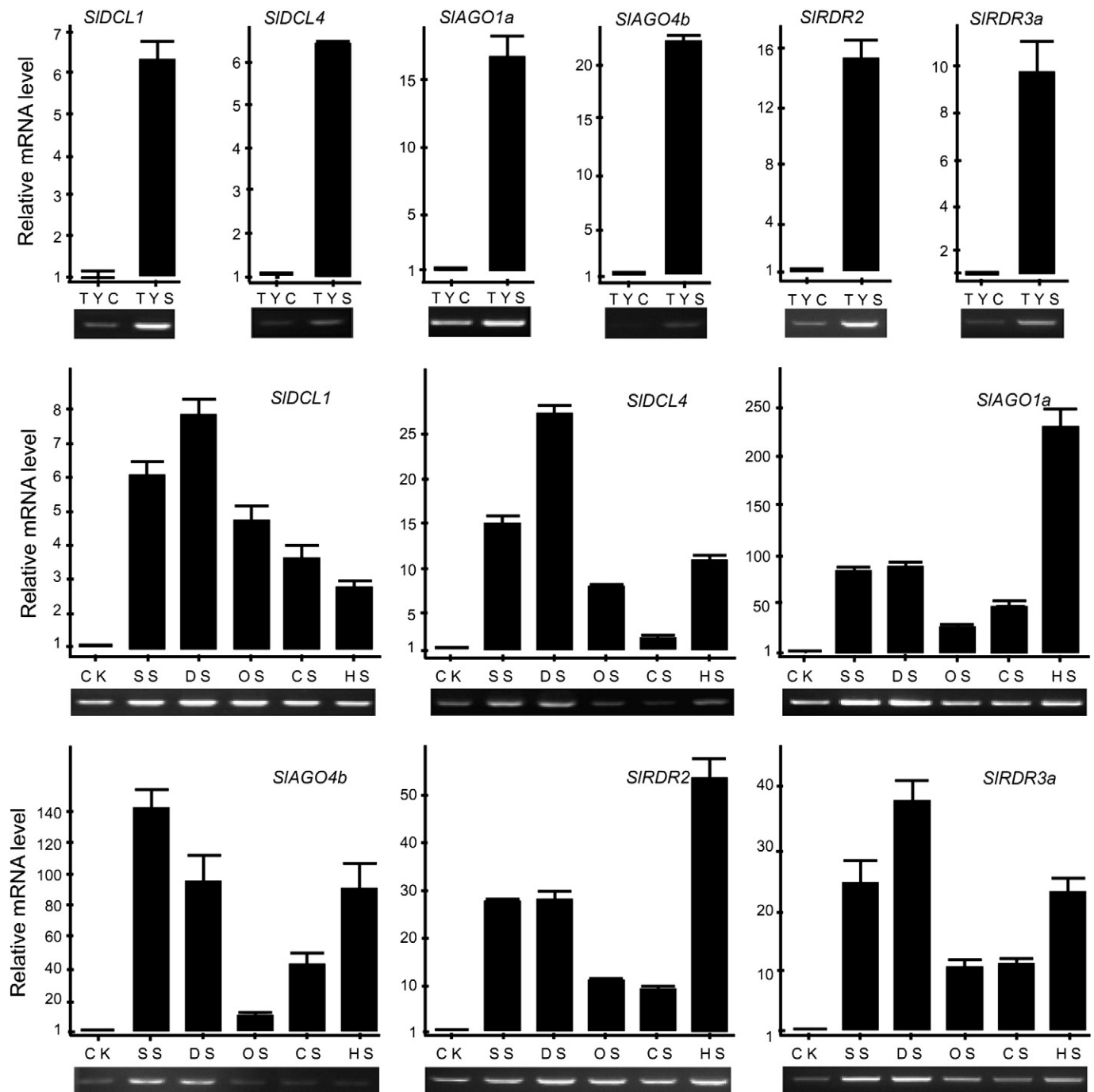


Fig. 6. Quantitative RT-PCR analysis of expression levels of *SIDCL1*, *SIDCL4*, *SIAGO1a*, *SIAGO4a*, *SIRDR2* and *SIRDR3a* under abiotic stress and viral infection treatments. The data represent raw expression values obtained from the analysis in the form of relative transcript abundance. Corresponding expression in each treatment obtained with semi-quantitative RT-PCR is indicated below the x-axis. The expression levels are shown compared to the respective control (TYC or CK). Error bars represent the standard error.

approach for tomato and viral genome methylation. *AtRDR6* acts in many RNA silencing pathways whose functions include defense against transgenes or viruses (Curaba and Chen, 2008; Vaistij and Jones, 2009). Expression patterns of *SIRDR6* showed clear differences in different tomato organs and in response to TYLCV infection and abiotic stresses. In tomato, significant expression of *SIRDR3a* occurs in reproductive organs and during stress responses.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at [doi:10.1016/j.gene.2012.02.009](https://doi.org/10.1016/j.gene.2012.02.009).

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